

Characterizing and Identifying Influencers on Bluesky

A comparative study of graph-based and cascade-based metrics

Marie Yu-Barreau

Polytech Grenoble - Computer Science (INFO4)

University of Tsukuba - Laboratory of Professor Sho Tsugawa

The Host Laboratory

University of Tsukuba, Japan

Laboratory of Professor Sho Tsugawa - Social Network Laboratory.

Why this lab?

Prior work on estimating psychological states (e.g. depressive tendency) from social media activity, combining large-scale data analysis with real human phenomena.



筑波大学
ソーシャルネットワーク研究室
Social Network Lab., University of Tsukuba

Identifying Real Influencers: Why It Matters

Marketing

Precise Targeting

Finding the right partners
for your campaigns

Budget Optimization

Investing in the most
profitable profiles

Public Policy and Crisis Communication

Health campaigns and crisis alerts
sometimes depend on reaching real leaders

But what actually makes someone influential?

What is the main characteristic of an Influencer?

Different types :

1

Popularity

Follower count

2

Authority

Followed / cited by other
important accounts

3

Spreading power

Actual ability to produce
content that induce debate

What is the main characteristic of an Influencer?

Different types :

1

Popularity

Follower count

2

Authority

Followed / cited by other important accounts

3

Spreading power

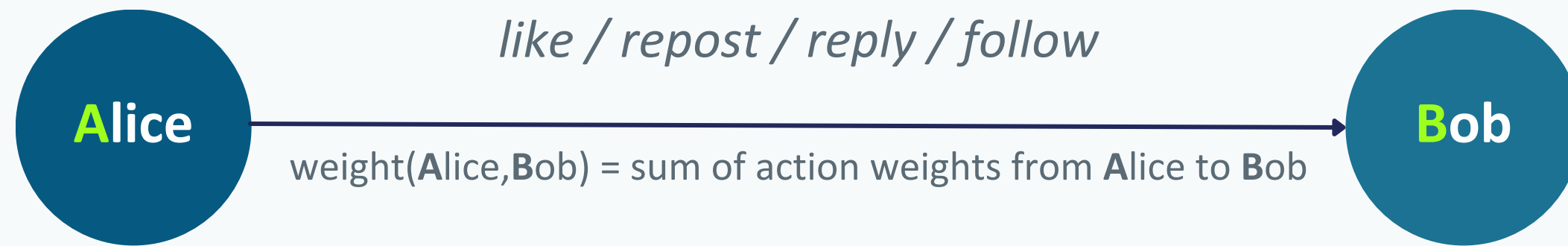
Actual ability to produce content that induce debate

Bluesky exposes its entire event stream (the firehose) publicly, an opportunity to reconstruct the full interaction graph of a real platform.

Three Main Objectives

- 1 Make data actionable: Build a pipeline reconstructing interaction graphs from raw Bluesky logs
- 2 Exploration: Apply and compare several graph-theoretic algorithms
- 3 Determine which classical structural algorithm best predicts our definition of spreading power, and with which setting
 - A Build an independent, dynamic ground truth of actual spreading power
 - B Quantitatively evaluate every algorithm with different settings against this ground truth

Building the Interaction Graphs



Example : For a weightmap **WM = (like: 1, repost: 50, reply: 15, follow: 30)**, `weight(Alice,Bob) = 5` means that **Alice** liked 5 posts posted by **Bob**.

Structural Algorithms Tested

PageRank

Authority / prestige of an account



Rang	Utilisateur	Score d'autorité
1	bsky.app	0.007077
2	dril.bsky.social	0.002616
3	safety.bsky.app	0.002393
4	xkcd.com	0.002060
5	ohkafuimykaful.bsky.social	0.001549

Ranking results of July 2023

Betweenness

Structural “bridge” role between communities



Rang	Utilisateur	Score Betweenness
1	shouldhavecat.bsky.social	0.019528
2	stovey.thequeer.house	0.014388
3	faineg.bsky.social	0.014104
4	catswithaura.bsky.social	0.010550
5	khanan.bsky.social	0.010459

K-Core

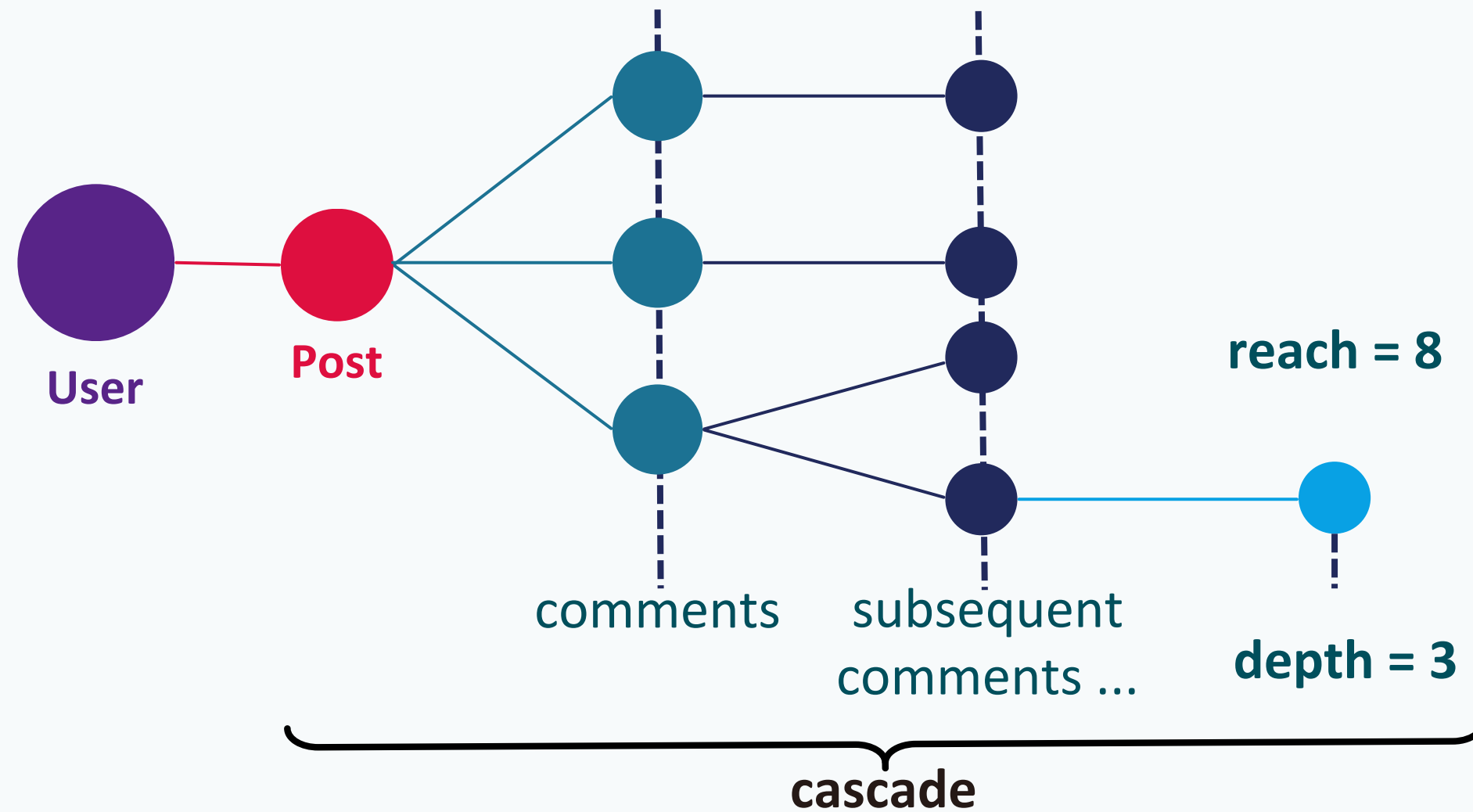
Density of the resilient core (with a prior filtering of weak links)



Rang	Utilisateur	K-Core (Shell)	Score Hybride	PageRank Brut	Betweenness Brut
1	ohkafuimykaful.bsky.social	61.0	0.2934	0.001549	0.0073
2	junlper.beer	61.0	0.1336	0.001450	0.0012
3	adamcerious.bsky.social	61.0	0.1170	0.001279	0.0011
4	darthbluesky.bsky.social	61.0	0.2367	0.000933	0.0068
5	estrogenempres.gay	61.0	0.2470	0.000865	0.0074

Which classical structural algorithm predict better our definition of spreading power? With which settings ?

A Dynamic Ground Truth: the Spreading Power using Replies



reach = **distinct** people reached
depth = longest transmission chain

Built from “Reply” event only (the only interaction letting us reconstruct a real chain of transmission between distinct people).

Spreading Power = (avg. reach × avg. depth²) × ln(1 + nb. cascades), computed on the top 25% cascades

Parameters in Settings Systematically Explored

- 1 Sub-network** Multiplex graph vs. Only-Like / Only-Follow / Only-Repост
- 2 Weighting scheme** Multiple weight models (weightmap) tested, including logarithmic weighting, and “no weighting”
- 3 Filtering threshold** Weak-link filtering: $X = 1, 2, 5, 30, 50, 100, 150$
- 4 Aggregation method** Single algorithm vs. Borda hybrid score

A “blind” graph (excluding replies) was used throughout to avoid leakage.

Main Result: Comparative Ranking over 18 Months of data

Method	Top-2000 overlap	Uncertainty	Spearman
PageRank — Like-only, unweighted	43.1%	± 3.7	0.370
PageRank — multiplex, unweighted	42.9%	± 3.9	0.386
Borda score — filtered multiplex	43.3%	± 4.7	0.307

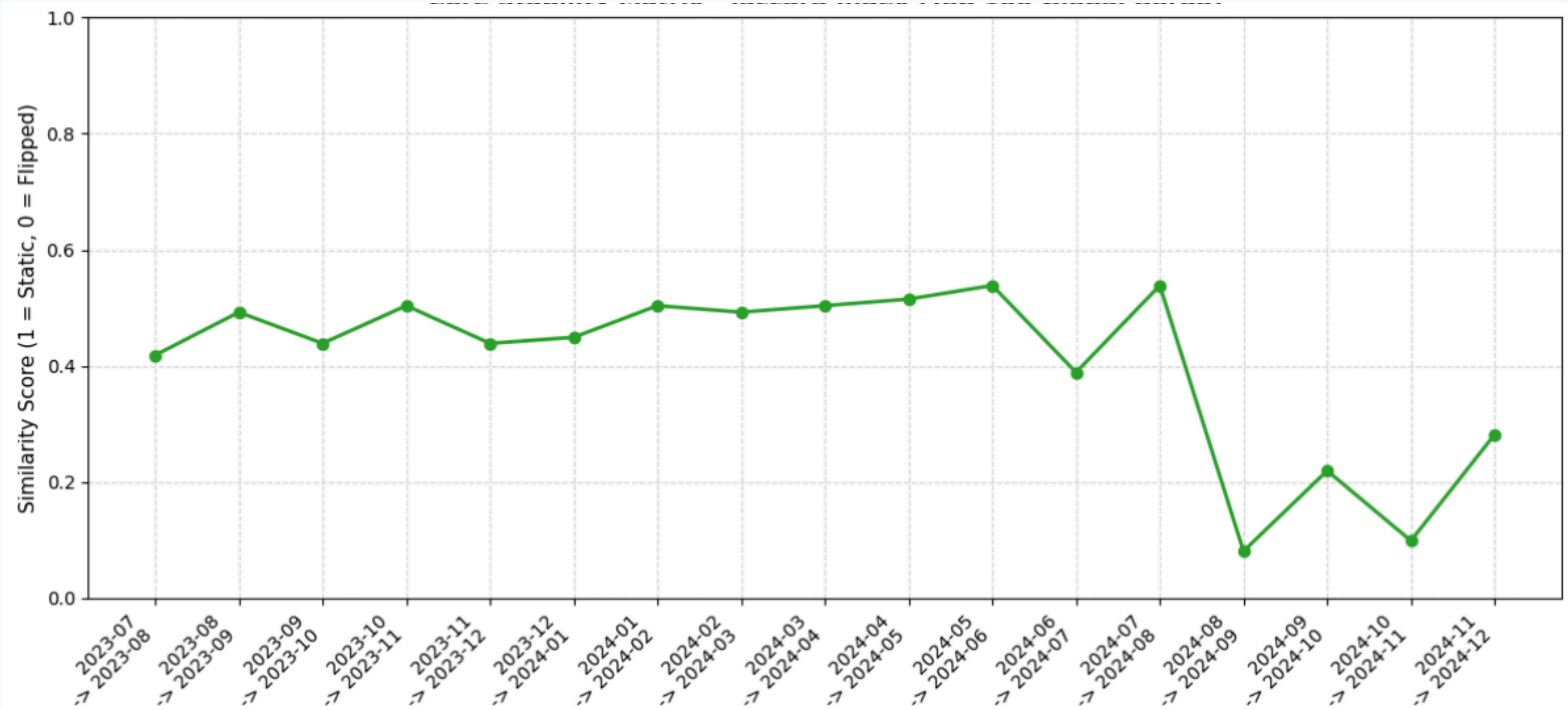
Unweighted PageRank on Likes-Only subgraph wins.

Borda's higher average overlap comes with a much higher month-to-month variance, and an even worse Spearman correlation.

Therefor, PageRank applied to the likes-only sub-network, without weighting or filtering, offers the best trade-off between simplicity, predictive power and stability over time.

Adding algorithmic complexity does not always translate into better result.

Temporal Stability of the Influencer Core (Top 100)

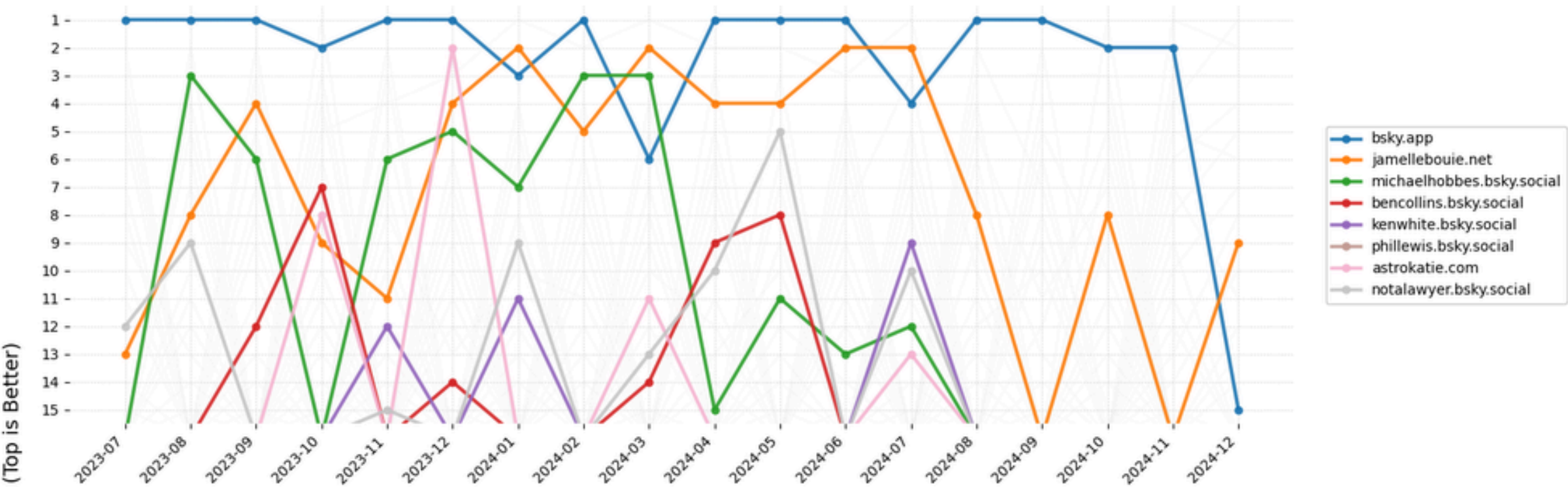


40–55% stable persistence, then a sharp drop from Aug. 2024 (coinciding with the **Nov. 2024 US election** surge).

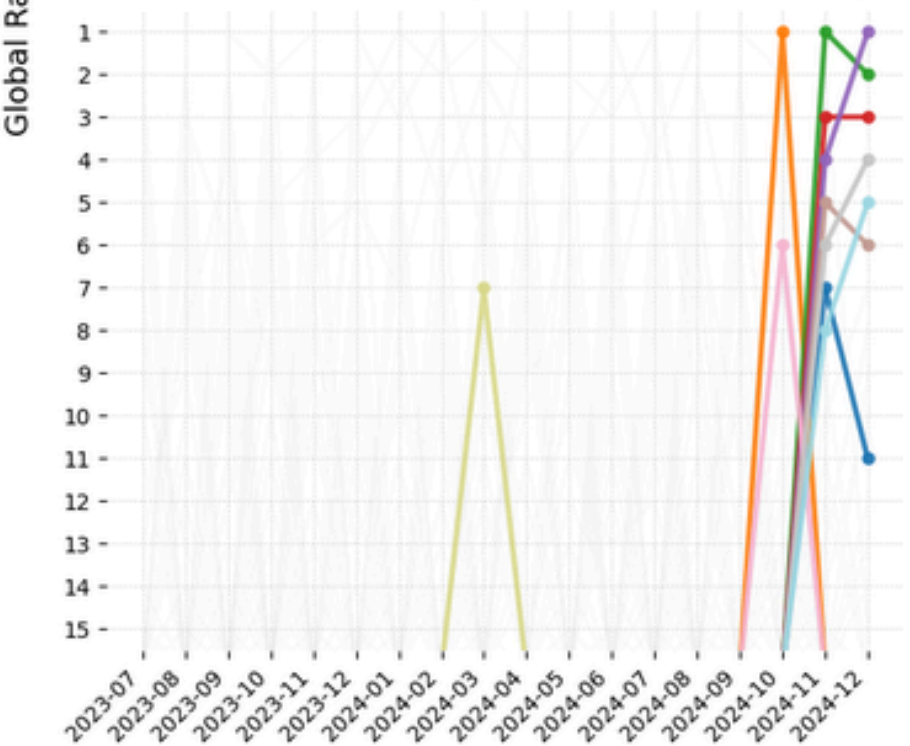
Typology of Influencer Trajectories

Global Leaderboard Evolution (Ground Truth - Top 15 Focus)

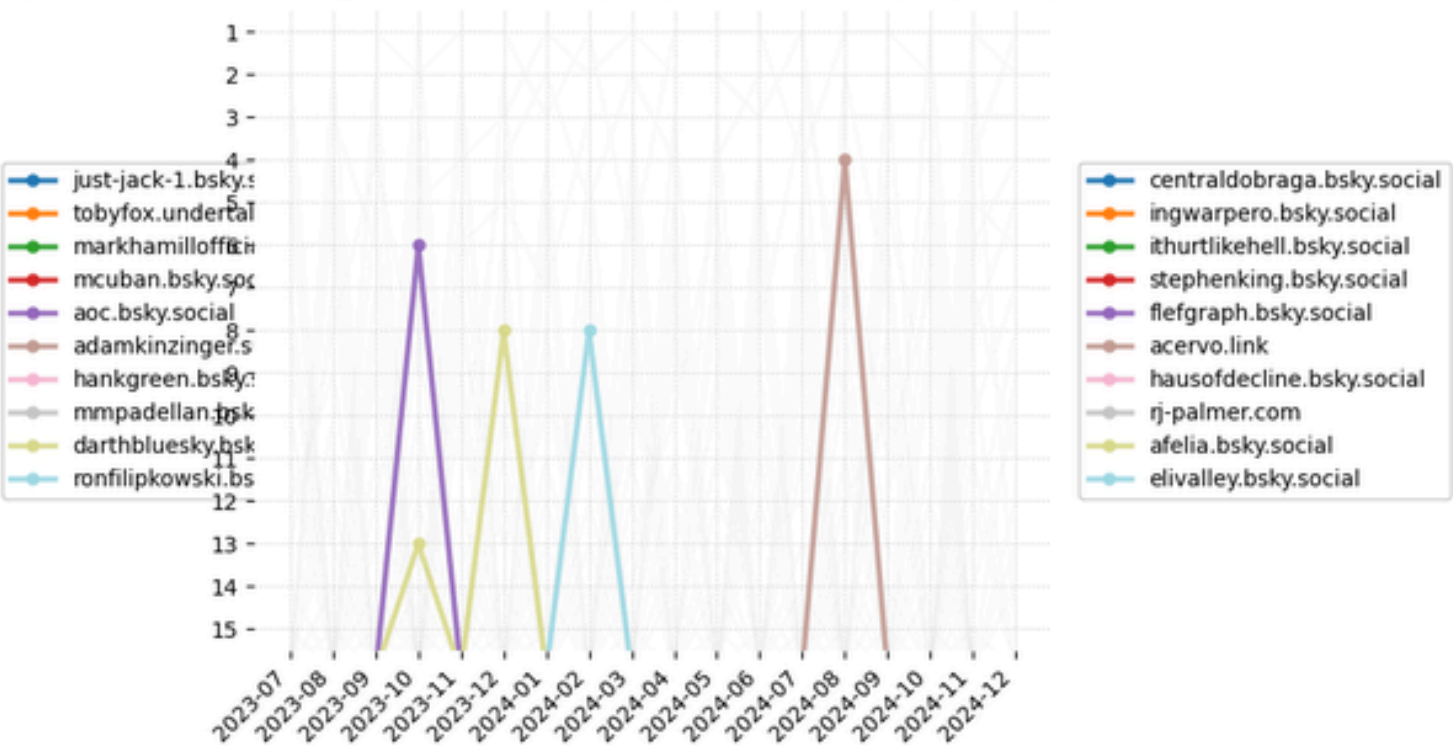
Pillars (Top constant, ≤ 1 absence) [n=8]



Steady Climbers (Inconnu → Top 15 & Maintien ≤ 500) [n=10]



Fading Leaders (Ex-Top 50, Chute ≥ 150 pl) [n=10]



Pillars

Present continuously in the Top 15 over all 18 months

Fading leaders

Reach the Top 15, then drop ≥150 places

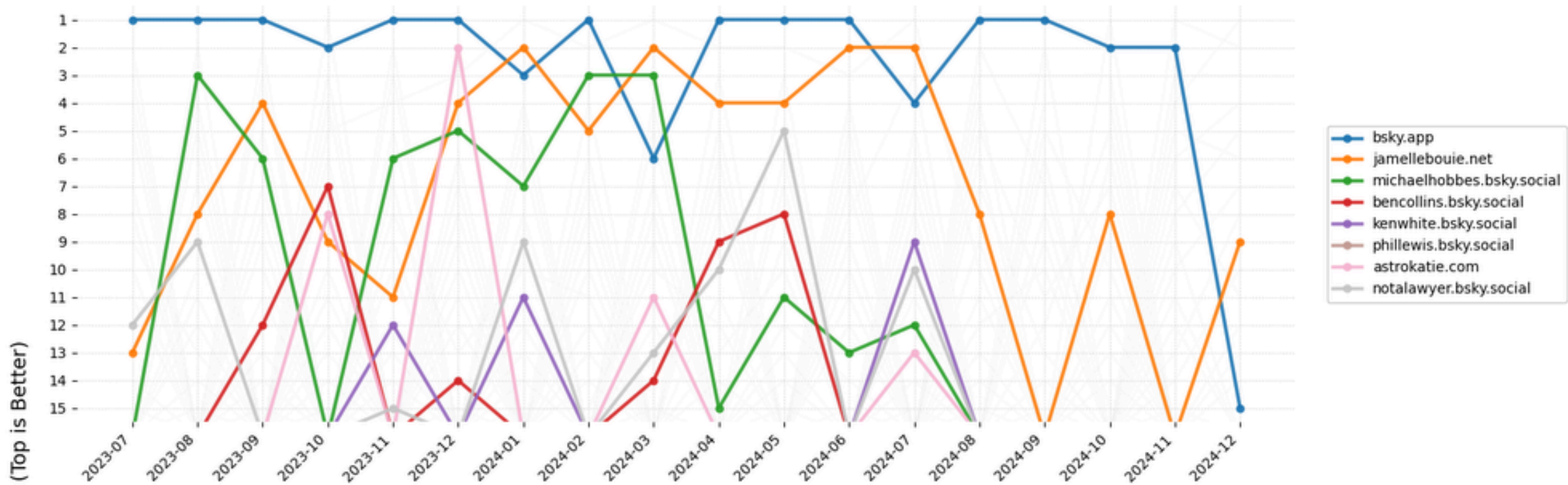
Steady climbers

Durably enter the Top 15 and remain in the Top 500

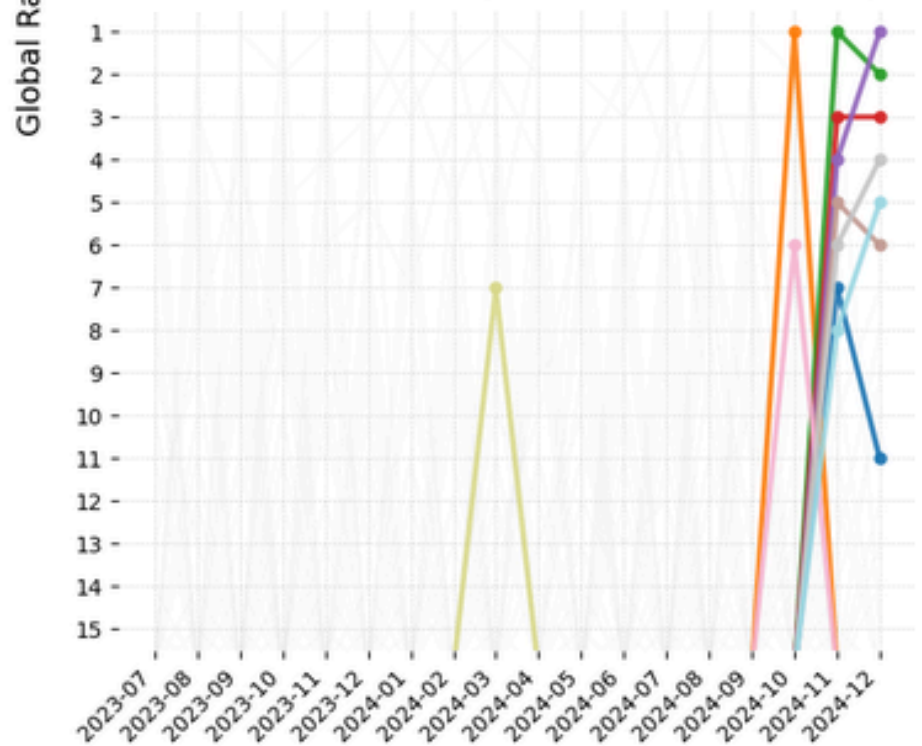
Typology of Influencer Trajectories

Global Leaderboard Evolution (Ground Truth - Top 15 Focus)

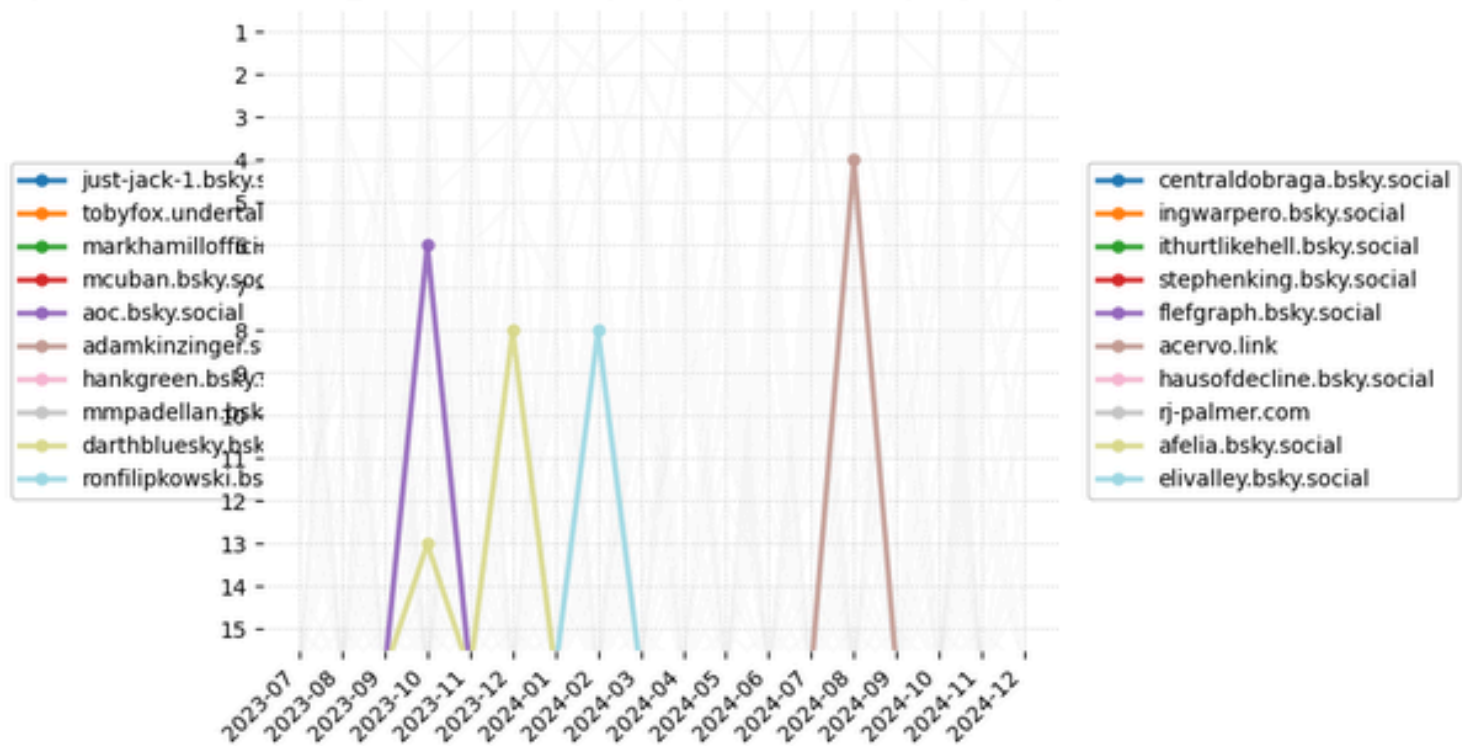
Pillars (Top constant, ≤ 1 absence) [n=8]



Steady Climbers (Inconnu → Top 15 & Maintien ≤ 500) [n=10]



Fading Leaders (Ex-Top 50, Chute ≥ 150 pl) [n=10]



Pillars

Present continuously in the Top 15 over all 18 months

Fading leaders

Reach the Top 15, then drop ≥150 places

Steady climbers

Durably enter the Top 15 and remain in the Top 500

Rising Stars

Start beyond the 2,000th spot then reach top 50 in less than two months

Two Secondary Objectives

- 1 Find the topics of Rising Star's posts on the moment of their rise
- 2 Trying to figure if we can predict future Rising Stars with structural algorithms

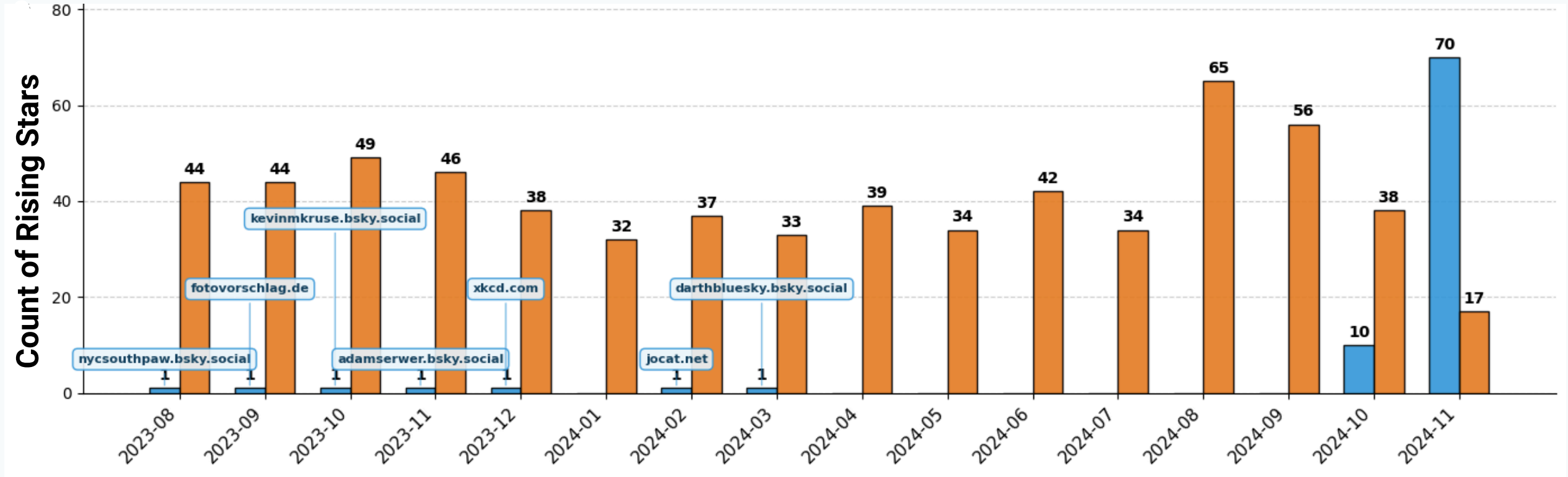
Lasting vs. Ephemeral Influence

Lasting influence

Accounts that reach top 50 and then maintain in top 500

Ephemeral influence (“pop & drop”)

Accounts that reach top 50 and then drop beneath top 500



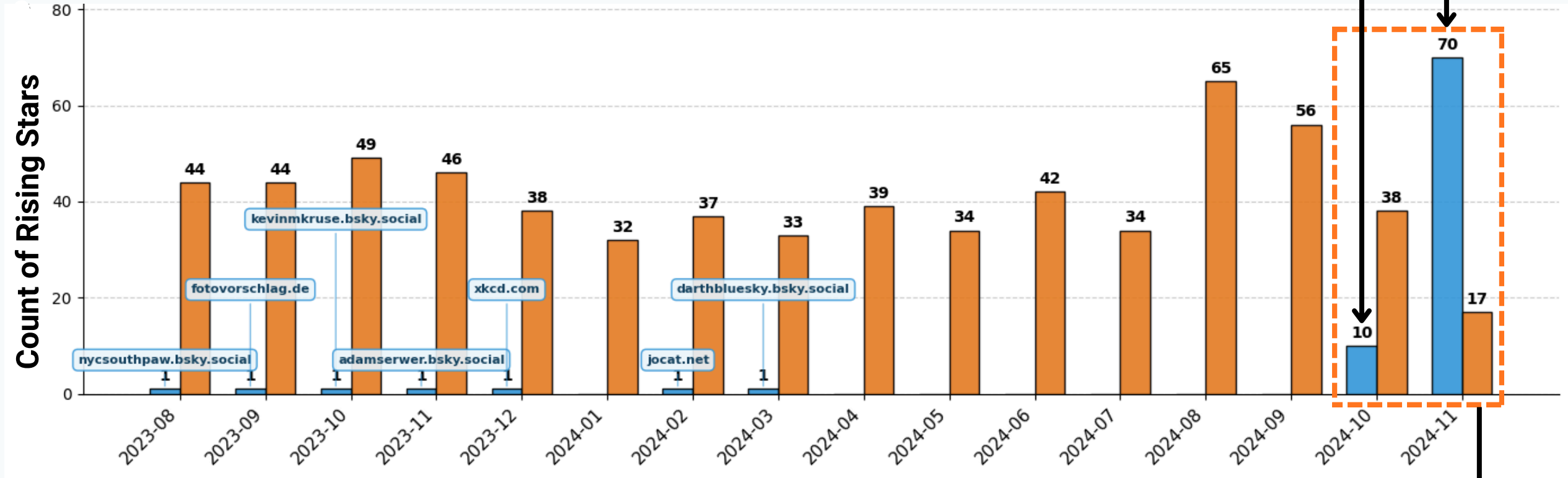
Lasting vs. Ephemeral Influence

Lasting influence

Accounts that reach top 50 and then maintain in top 500

Ephemeral influence (“pop & drop”)

Accounts that reach top 50 and then drop beneath top 500



Could become orange with more months of data

Remaining in the top 50 during 2-3 months is easier than during 18 months

RESULTS

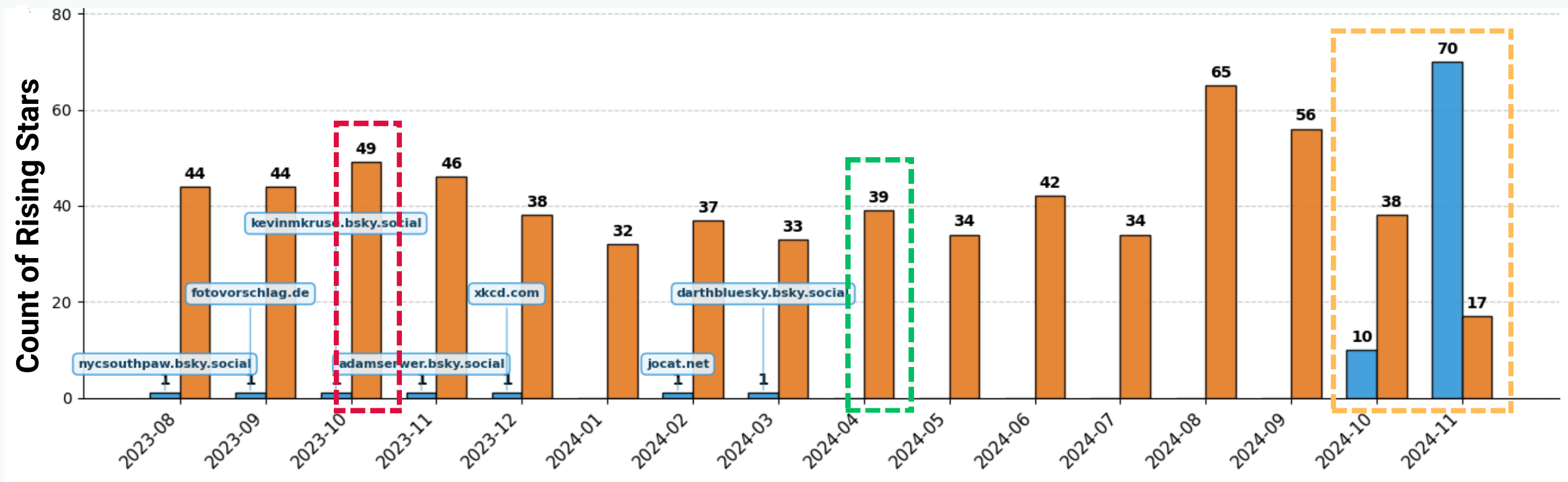
Lasting vs. Ephemeral Influence

Lasting influence

Accounts that reach top 50 and then maintain in top 500

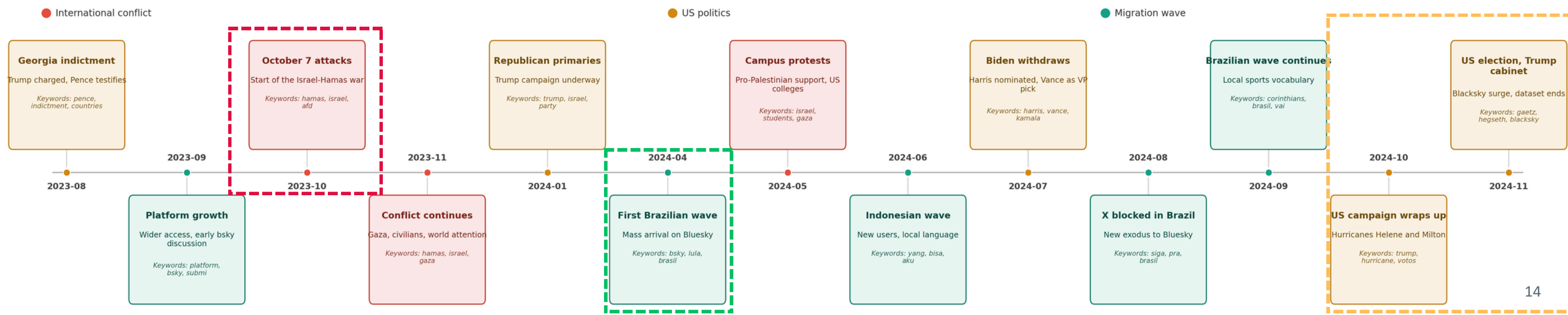
Ephemeral influence (“pop & drop”)

Accounts that reach top 50 and then drop beneath top 500



Ephemeral Influence ↔ Real-World Events

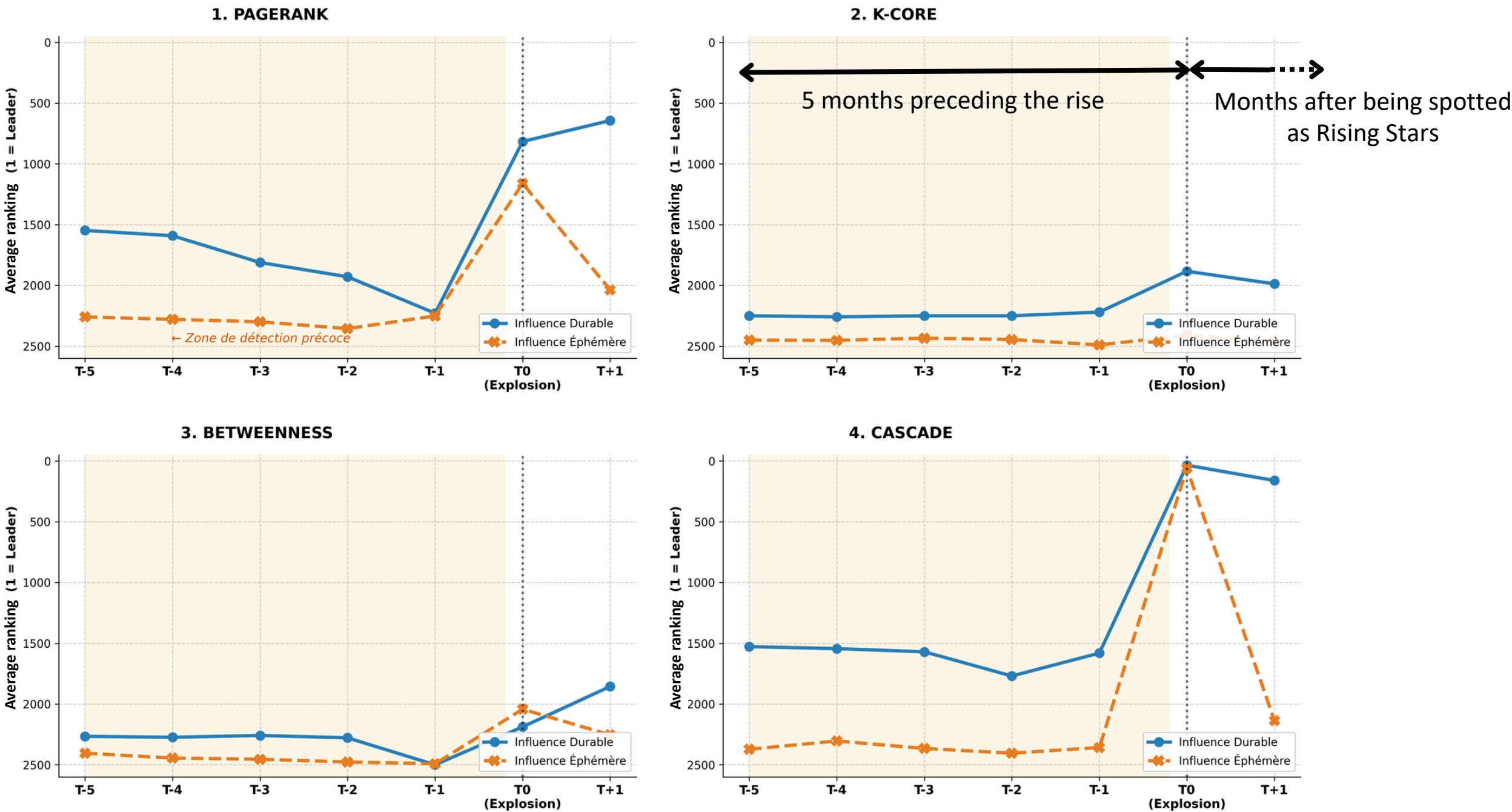
Timeline of events linked to influence spikes on Bluesky, Aug 2023 to Nov 2024



Trying to see if Rising Stars are predicatble

- Lasting influence Rising Stars
- Ephemeral influence Rising Stars

Evaluation of algorithm predictive power: Rising star for October 2024



Structural algorithms only react at the moment of explosion, apparently there is no early prediction capability with the tools tested.

Toward Early Prediction

Structural algorithms only react at the moment of explosion. A finer approach:

- 1 Machine learning on precursor structural indicators
- 2 Semantic content matched against currently trending topics

Goal: shift from structural characterization to genuine early prediction.

Specialty Competency

Designing Data, Language, Image and Human-Centered Systems

- **Collaborating with end users** Regular exchanges with Professor Tsugawa
- **Choosing suitable algorithms** Selecting & comparing PageRank, K-Core, Betweenness
- **Structuring the data** Building a pipeline reconstructing interaction graph into parquet edge lists
- **Validating through evaluation** Automated test bench vs. ground truth (overlap, Spearman)
- **Ethics / GDPR compliance** Public firehose data, Only worked on quantitative data, Only displayed pseudo (no personal information such as age, sex, opinion...)

Personal Takeaways

What I learned

- Iterating on a research question
- Accepting that an initial approach can be wrong
- How graph theory can study a real social phenomenon
- Adapting to a new living & working environment in Japan

Thank you !

With thanks to:

Professor Sho Tsugawa — supervision throughout the research

Yuina Takahashi — integration into the laboratory

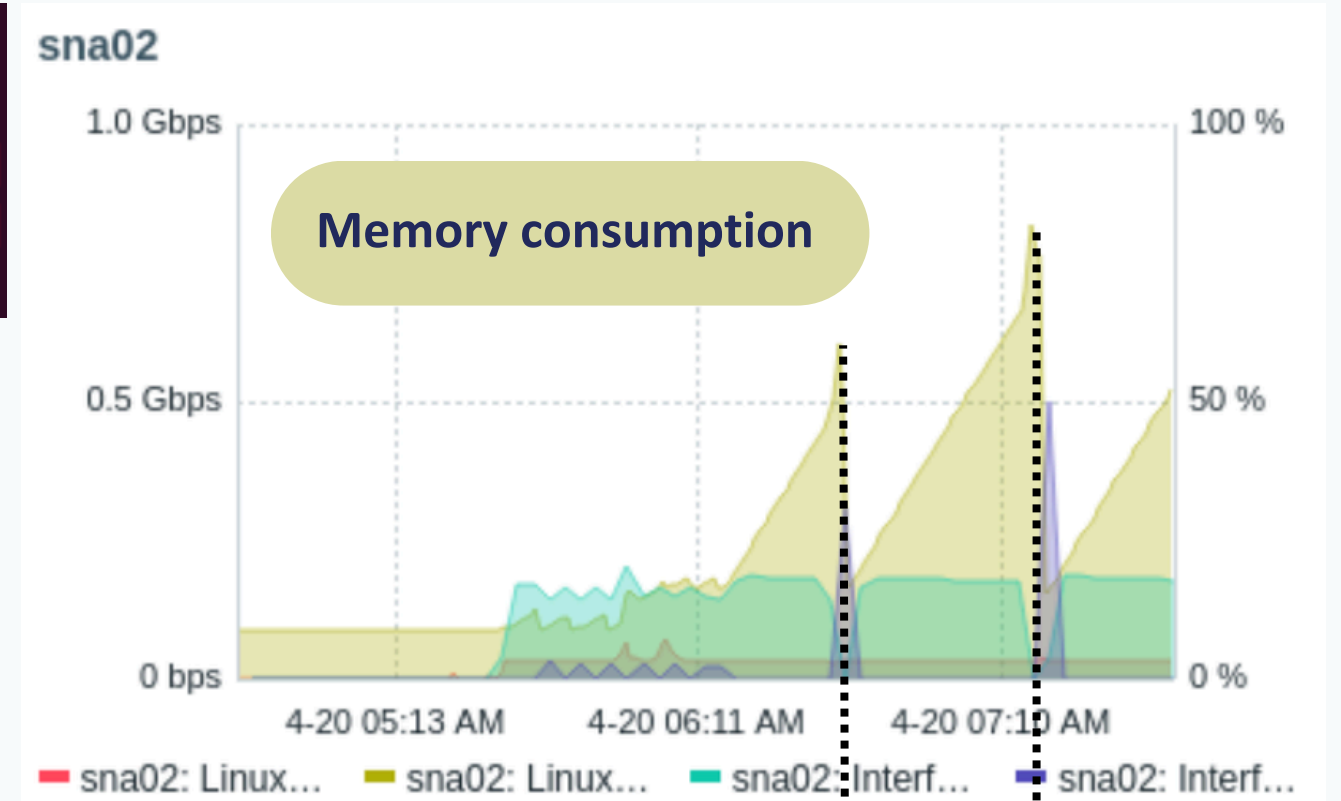
Yuto Tamura — technical assistance

M. Blanch — academic supervision at Polytech Grenoble

Scaling to 18 Months of Data

The platform's data volume grew rapidly (peaking around the Nov. 2024 US election), requiring successive infrastructure rebuilds:

```
(venv) MarieYU@now23:~/InfluencersOnBluesky$ python 2024_12_ExtractionScreen.py
--- Processing 2024-12 ---
0%|                                     | 2/1488 [00:16<3:24:53, 8.27s/it]
56%|██████████████████████████████████| 835/1488 [2:36:17<12:43:21, 70.14s/it]
Killed
```



Solutions :

1

Pandas → DuckDB

Memory-safe aggregation with disk-spill support
→ Eliminated the RAM saturation crashes that killed the pipeline mid-run

2

Brandes approximation

Adaptive sampling for Betweenness, bounded 5% error margin
→ Made centrality computation tractable on million-node graphs, at negligible accuracy cost

3

Cascade noise filtering

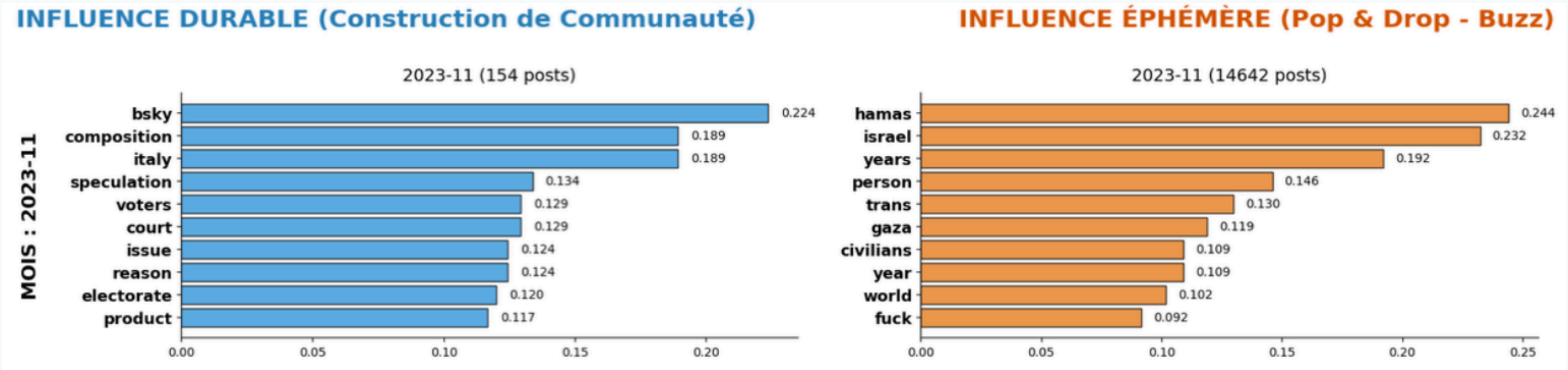
Discard threads with ≤ 3 messages before reconstruction
→ Cut processing volume by 85–90%, with no loss of signal on genuine influence cascades

Transverse Competency

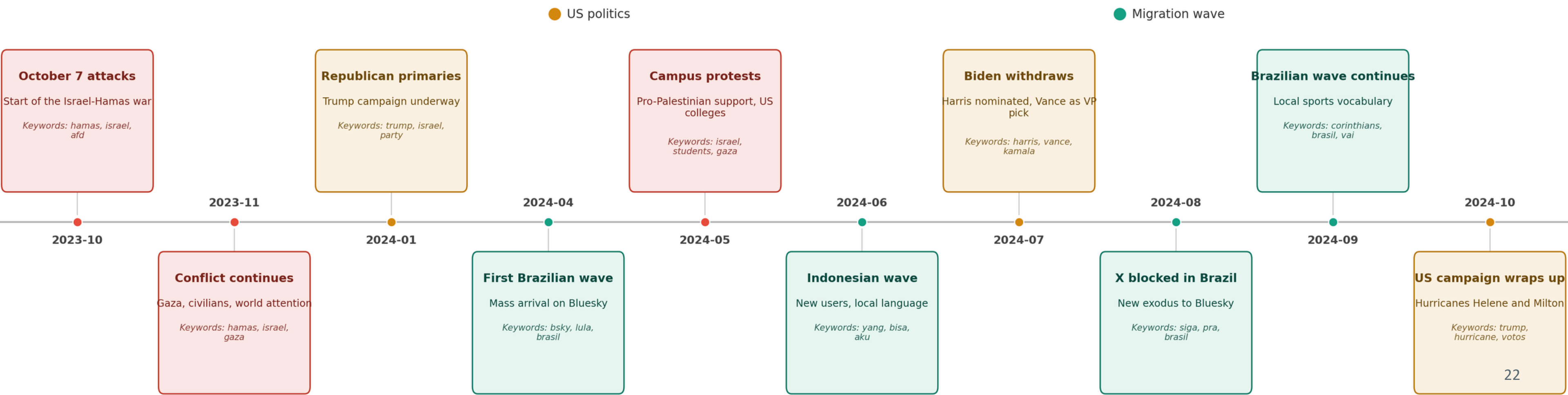
Engineer who collaborates & shares results

- Weekly logbook & meetings with Prof. Tsugawa
- Argued every methodological choice (thresholds, model)
- Own choices within the lab's collective framework
- Optimized compute footprint + eco-friendly local mobility

Semantic Analysis (TF-IDF)



Timeline of events linked to influence spikes on Bluesky, Aug 2023 to Nov 2024



Comparison of algorithms using different settings against the ground truth

DÉMARRAGE DE L'ANALYSE DU MOIS : 2024-01									
MULTIPLEX Betweenness : Degré >=1000 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	45.1%	0.36	25.9%	0.25	13.2%	0.14	38.7%	0.272	
2	43.1%	0.36	28.0%	0.26	30.6%	0.19	46.3%	0.291	
5	41.1%	0.37	28.7%	0.27	39.0%	0.25	47.1%	0.317	
30	40.1%	0.37	32.5%	0.28	33.9%	0.27	48.1%	0.339	
50	41.5%	0.35	34.1%	0.28	23.9%	0.28	47.8%	0.330	
150	25.2%	0.23	26.8%	0.23	1.3%	0.11	35.5%	0.242	

ONLY_LIKE Betweenness : Degré >=800 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	45.3%	0.34	23.8%	0.25	17.2%	0.16	39.3%	0.268	
2	41.2%	0.26	26.5%	0.22	36.3%	0.25	42.3%	0.246	
5	36.2%	0.16	25.7%	0.14	14.3%	0.30	36.0%	0.158	
30	24.9%	0.04	14.8%	0.03	4.4%	0.00	24.0%	0.032	
50	21.6%	0.02	14.5%	0.01	4.4%	0.00	21.8%	0.018	
150	16.9%	0.05	14.2%	0.03	4.4%	0.00	17.0%	0.039	

ONLY_FOLLOW Betweenness : Degré >=50 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	28.6%	0.36	2.2%	0.28	10.5%	0.24	30.1%	0.332	
2	23.9%	0.32	17.5%	0.30	2.1%	0.08	26.5%	0.324	
5	9.7%	0.20	9.6%	0.19	1.7%	0.00	9.6%	0.196	
30	1.8%	0.03	1.8%	0.03	1.7%	0.00	1.8%	0.032	
50	1.7%	0.04	1.7%	0.04	1.7%	0.00	1.7%	0.039	
150	1.7%	0.04	1.7%	0.04	1.7%	0.00	1.7%	0.039	

ONLY_REPOST Betweenness : Degré >=30 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	41.2%	0.35	34.0%	0.28	15.5%	0.15	45.5%	0.283	
2	29.0%	0.29	29.0%	0.25	14.8%	0.14	39.6%	0.270	
5	19.7%	0.18	22.8%	0.20	17.2%	0.15	27.5%	0.198	
30	11.6%	0.11	14.2%	0.11	4.3%	0.07	11.9%	0.110	
50	13.1%	0.11	11.7%	0.11	3.9%	0.05	12.2%	0.111	
150	7.2%	0.05	7.0%	0.06	3.9%	0.00	7.0%	0.057	

DÉMARRAGE DE L'ANALYSE DU MOIS : 2024-02									
MULTIPLEX Betweenness : Degré >=1000 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	35.6%	0.36	26.1%	0.28	10.5%	0.11	37.9%	0.284	
2	33.3%	0.34	2.2%	0.27	16.7%	0.20	37.9%	0.307	
5	31.8%	0.34	0.6%	0.28	20.0%	0.27	32.6%	0.320	
30	30.8%	0.35	0.5%	0.29	19.7%	0.26	30.2%	0.329	
50	35.1%	0.35	31.1%	0.28	26.6%	0.28	47.4%	0.327	
150	23.2%	0.23	26.7%	0.23	1.6%	0.12	32.5%	0.230	

ONLY_LIKE Betweenness : Degré >=800 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	36.4%	0.35	24.6%	0.28	13.7%	0.15	39.2%	0.286	
2	33.1%	0.27	26.8%	0.24	30.1%	0.26	42.9%	0.266	
5	31.1%	0.17	23.6%	0.16	15.2%	0.31	36.0%	0.175	
30	23.0%	0.03	16.4%	0.02	4.9%	0.08	24.5%	0.018	
50	20.5%	0.01	16.2%	0.01	4.8%	0.00	22.0%	0.006	
150	14.9%	0.01	14.1%	0.00	4.8%	0.00	14.8%	0.007	

ONLY_FOLLOW Betweenness : Degré >=50 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	17.7%	0.33	0.5%	0.28	5.8%	0.30	4.9%	0.314	
2	14.0%	0.28	4.4%	0.30	5.0%	0.16	11.5%	0.298	
5	8.5%	0.25	10.1%	0.25	1.6%	0.00	8.5%	0.253	
30	3.1%	0.10	3.0%	0.10	1.6%	0.00	3.0%	0.099	
50	1.8%	0.03	1.8%	0.04	1.6%	0.00	1.8%	0.045	
150	1.7%	0.04	1.7%	0.04	1.6%	0.00	1.7%	0.039	

ONLY_REPOST Betweenness : Degré >=30 (Max 15000 nœuds)									
Seuil	PageRank		K-Core		Betweenness		BORDA		
	Ov	Sp	Ov	Sp	Ov	Sp	Ov	Sp	
1	36.4%	0.36	31.1%	0.29	14.3%	0.16	43.9%	0.305	
2	26.6%	0.29	30.0%	0.26	10.9%	0.18	35.6%	0.267	
5	18.7%	0.17	21.9%	0.20	16.8%	0.18	26.2%	0.190	
30	10.6%	0.08	14.2%	0.09	4.8%	0.04	10.5%	0.079	
50	10.9%	0.06	11.5%	0.06	4.4%	0.00	10.6%	0.057	
150	8.2%	0.03	8.2%	0.03	4.4%	0.00	8.2%	0.028	

Comparison of algorithms using different settings against the ground truth (over the 18months)

OVERLAP TOP 2000 (%) - RÉSEAU : LIKE						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	47.85	46.30	37.35	31.55	40.35	40.40
2023-08	41.90	39.65	30.00	23.10	33.60	33.40
2023-09	38.35	37.80	25.80	19.50	30.90	31.00
2023-10	39.05	37.00	22.35	16.35	28.65	28.45
2023-11	40.95	39.60	22.20	21.50	29.55	29.80
2023-12	41.65	40.10	22.75	18.00	29.70	29.50
2024-01	45.30	41.45	25.45	25.45	34.80	35.40
2024-02	36.40	33.50	20.95	26.00	34.70	34.65
2024-03	40.55	36.85	24.30	27.40	36.40	36.10
2024-04	44.35	39.25	24.85	26.90	36.60	36.00
2024-05	43.45	39.70	25.60	28.55	36.90	36.85
2024-06	43.90	39.90	24.05	26.05	33.70	34.35
2024-07	45.95	41.25	27.10	30.90	38.10	39.35
2024-08	46.10	42.85	20.55	24.25	29.60	30.00
2024-09	44.65	41.20	17.75	13.45	28.25	28.05
2024-10	38.15	37.00	16.60	17.60	31.00	30.50
2024-11	46.75	42.70	15.60	13.10	30.25	29.60
2024-12	50.55	44.85	15.90	20.80	35.20	35.85
=> MOYENNE GLOBALE DU RÉSEAU LIKE :						
PR (Unweighted)	43.10					
PR (Weighted)	40.05					
Betweenness	23.29					
K-Core	22.80					
Borda (Unw)	33.24					
Borda (W)	33.29					

OVERLAP TOP 2000 (%) - RÉSEAU : REPOST						
Month	PR (Unweighted)	PR (Weighted)	Betweenness	K-Core	Borda (Unw)	Borda (W)
2023-07	43.25	40.65	34.95	30.90	40.50	40.40
2023-08	39.20	37.15	28.50	25.60	35.15	35.15
2023-09	34.80	33.15	24.35	22.50	32.45	32.40
2023-10	35.30	34.60	21.95	24.70	33.20	33.00
2023-11	37.00	33.95	23.50	25.60	34.25	34.15
2023-12	34.50	33.65	23.75	22.95	35.85	35.45
2024-01	41.25	37.50	25.85	29.00	39.10	38.75
2024-02	36.35	33.75	21.55	30.05	36.30	35.90
2024-03	40.00	36.55	24.75	30.65	40.20	39.55
2024-04	42.90	39.55	22.30	29.00	39.50	39.45
2024-05	41.45	38.55	24.15	28.40	39.55	39.25
2024-06	43.15	40.05	23.30	29.80	36.80	36.80
2024-07	44.20	40.60	26.20	30.35	41.60	41.60
2024-08	41.45	38.50	21.00	23.70	32.85	32.70
2024-09	37.15	35.95	15.90	12.10	30.35	30.65
2024-10	32.50	30.85	13.15	17.55	31.50	31.15
2024-11	38.90	36.20	14.80	21.15	36.20	36.10
2024-12	44.40	40.55	16.75	22.20	41.00	40.80
=> MOYENNE GLOBALE DU RÉSEAU REPOST :						
PR (Unweighted)	39.32					
PR (Weighted)	36.76					
Betweenness	22.59					
K-Core	25.34					
Borda (Unw)	36.46					
Borda (W)	36.29					

Cascade example

Propagation Topology of “bsky.app” best cascade
Reach : 473 | Depth : 6

